Shear Banding & Amourphous Materials

Presented by Kondwani Phiri

CONTENT

01 BACKGROUND & PURPOSE

02 SYNPOSIS/PROCESS

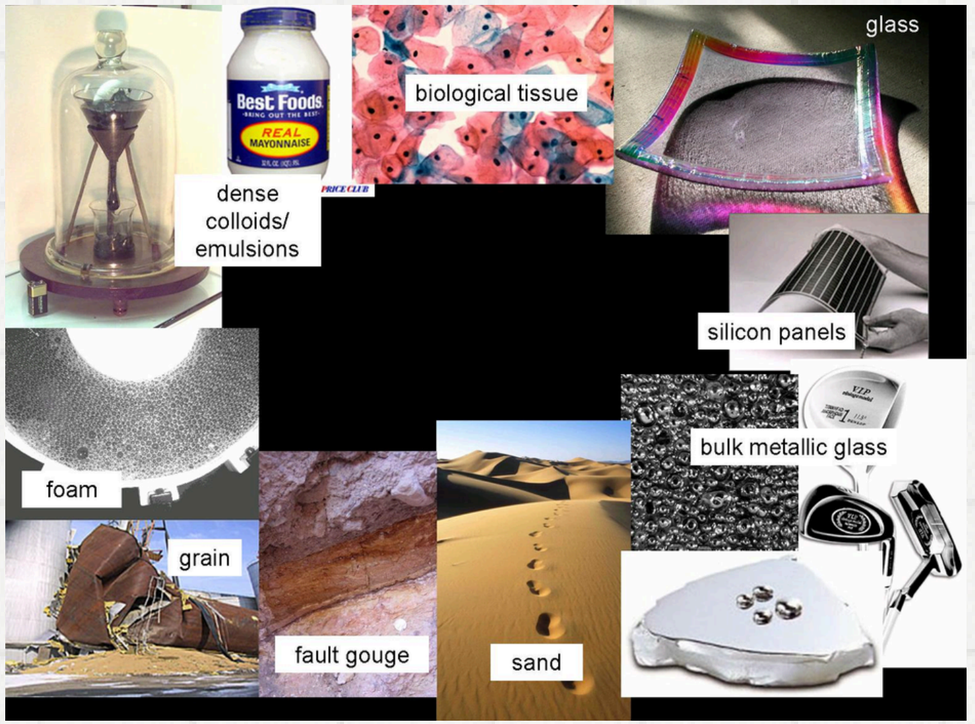
03 APPLICATIONS

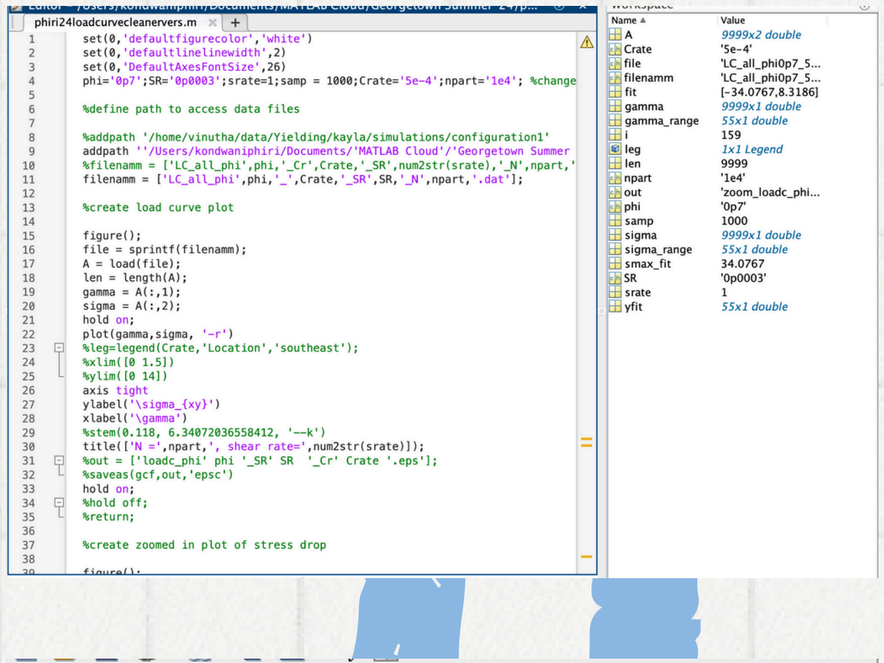
04 RESULTS

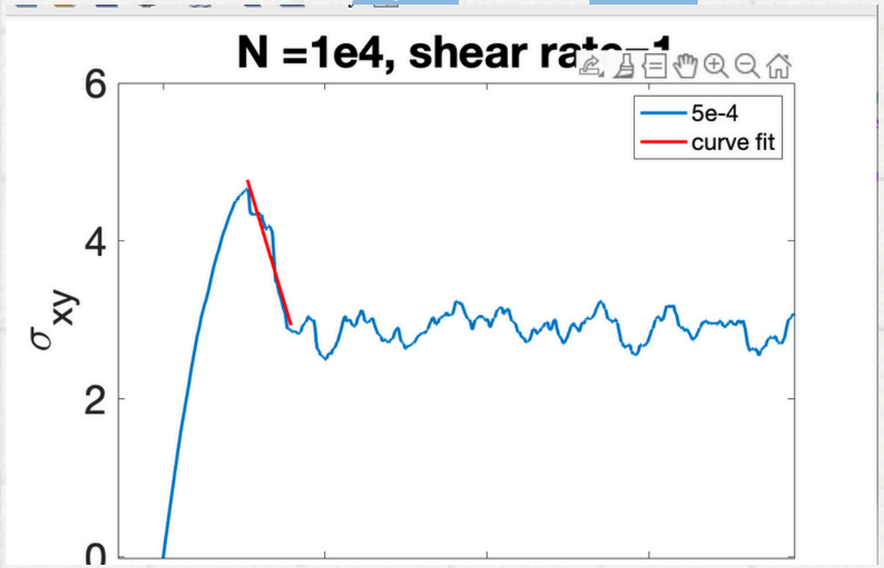
05 INSIGHTS + FUTURE

Background & Purpose

- Amorphous Material
- Shear Bandings
  - Why it matters
- Research and what it does
  - Landmark Paper published in 1976 by Mahmudur Rahman and Woodcock (Bangladesh University of Health Sciences)







Process

- Read Research / Data
  - Get a sense of material
  - Be able to explain to a 13 year old
- Matlab & QtGrace Plotting
  - Laying out data and writing differences between what i've found versus what I though expected to happen
- Comparison with Previous Data
  - (later in data analysis)

Applications

Amorphous Materials and shear banding have many applications throughout Physics, Chemistry & multiple disciplines of STEM. As a mechanical engineer myself, I tried to adapt three applications that I believed would be most common to find working as a modern engineer.

01. Product Design



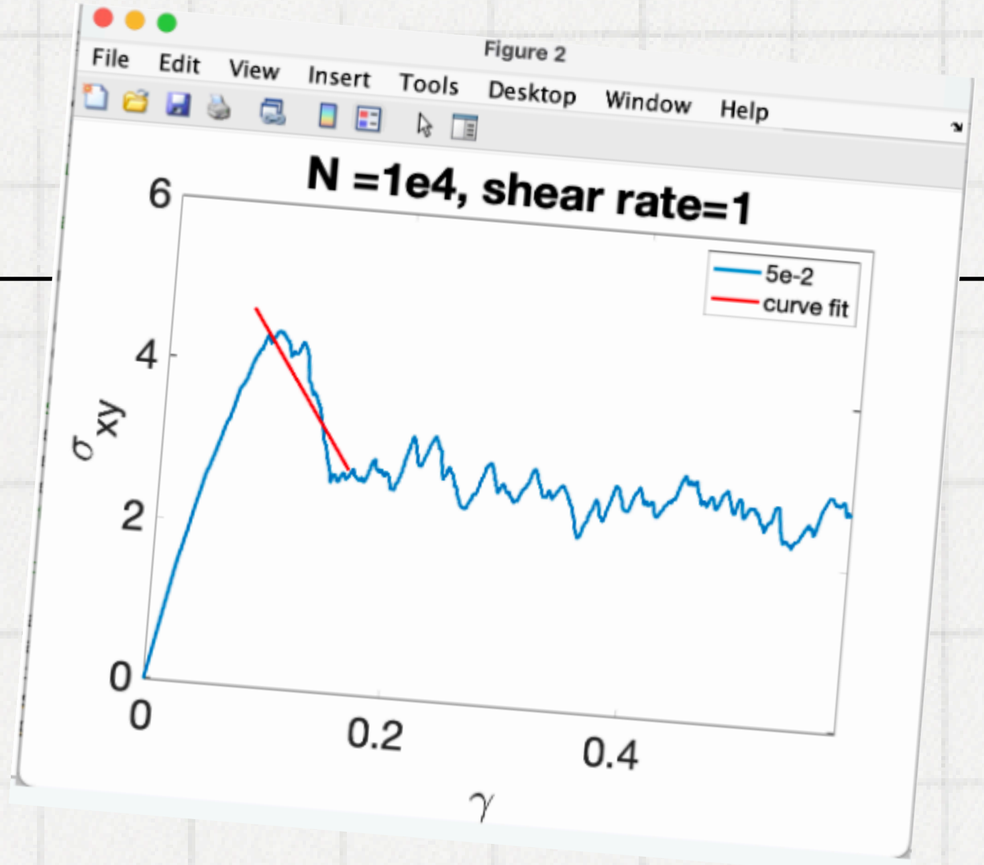


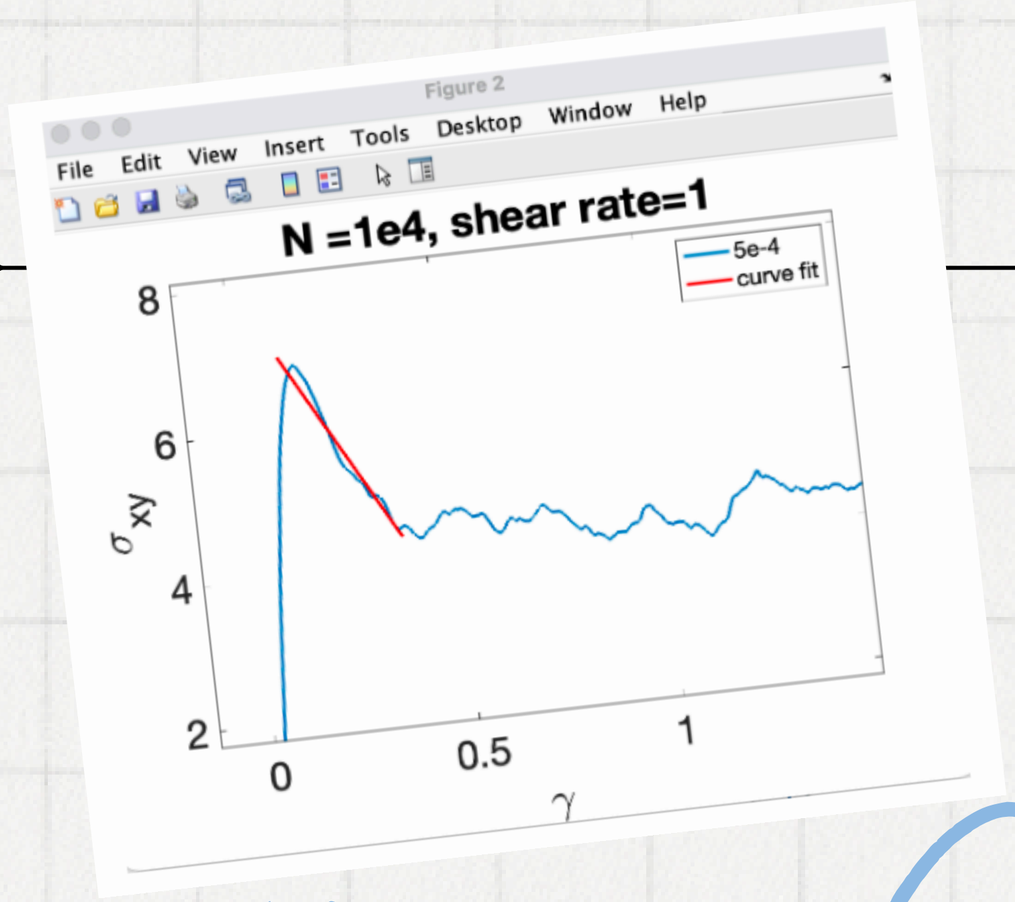


02. Petrol & Chemical Engineering

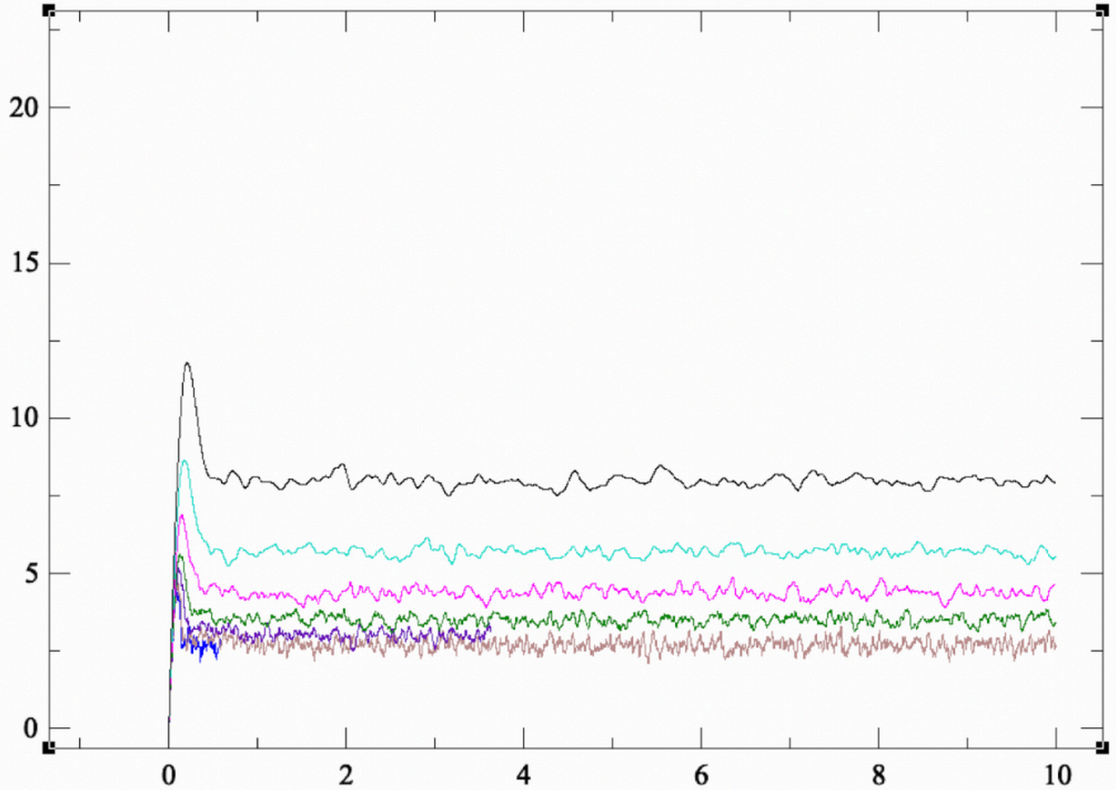
03. **CIVIL ENGINEERING AND BUILDING SCIENCE**

Main Results

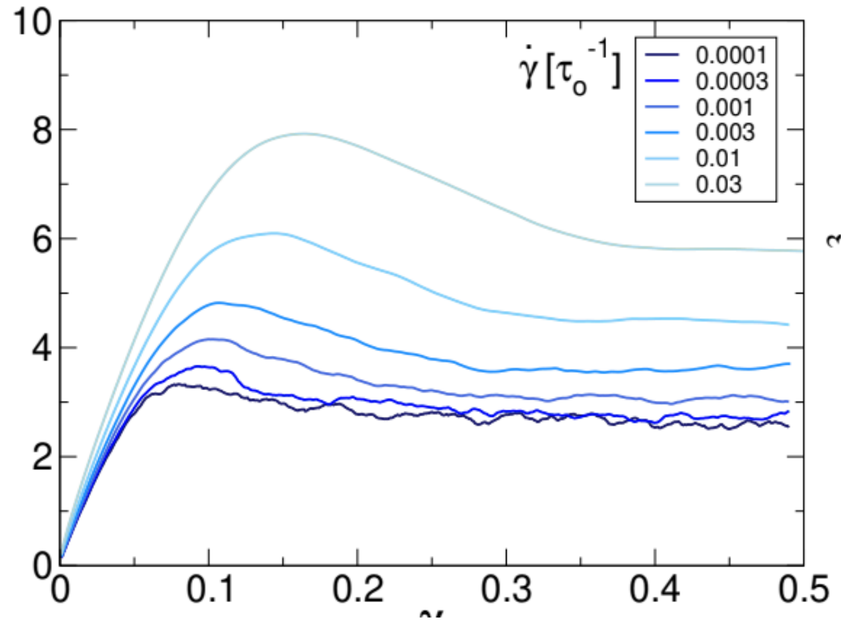




CONFIGURATION 1



Load curves like those shown in Figure 11, were made and examined for each sample preparation (depending on the cooling rate), and for each imposed shear rate, $\dot{\gamma}[\tau_0^{-1}]$. Figure 11a shows the load curves for the least aged sample, which was prepared at the highest cooling rate of $5 \times 10^{-2}[\varepsilon/k_B\tau_0]$. As the shear rate $\dot{\gamma}[\tau_0^{-1}]$ is increased, the maximum value of shear stress $\sigma[\varepsilon/a^3]$ that the system reaches increases to a larger peak; the stress drop remains relatively gradual for each of the imposed shear rates, as the samples gradually transition to flow over a long strain range. Therefore, as outlined in section 2.2, each of these curves correspond to a relatively low S_{max} value, suggesting that a low S_{max} value is characteristic for very poorly aged samples.



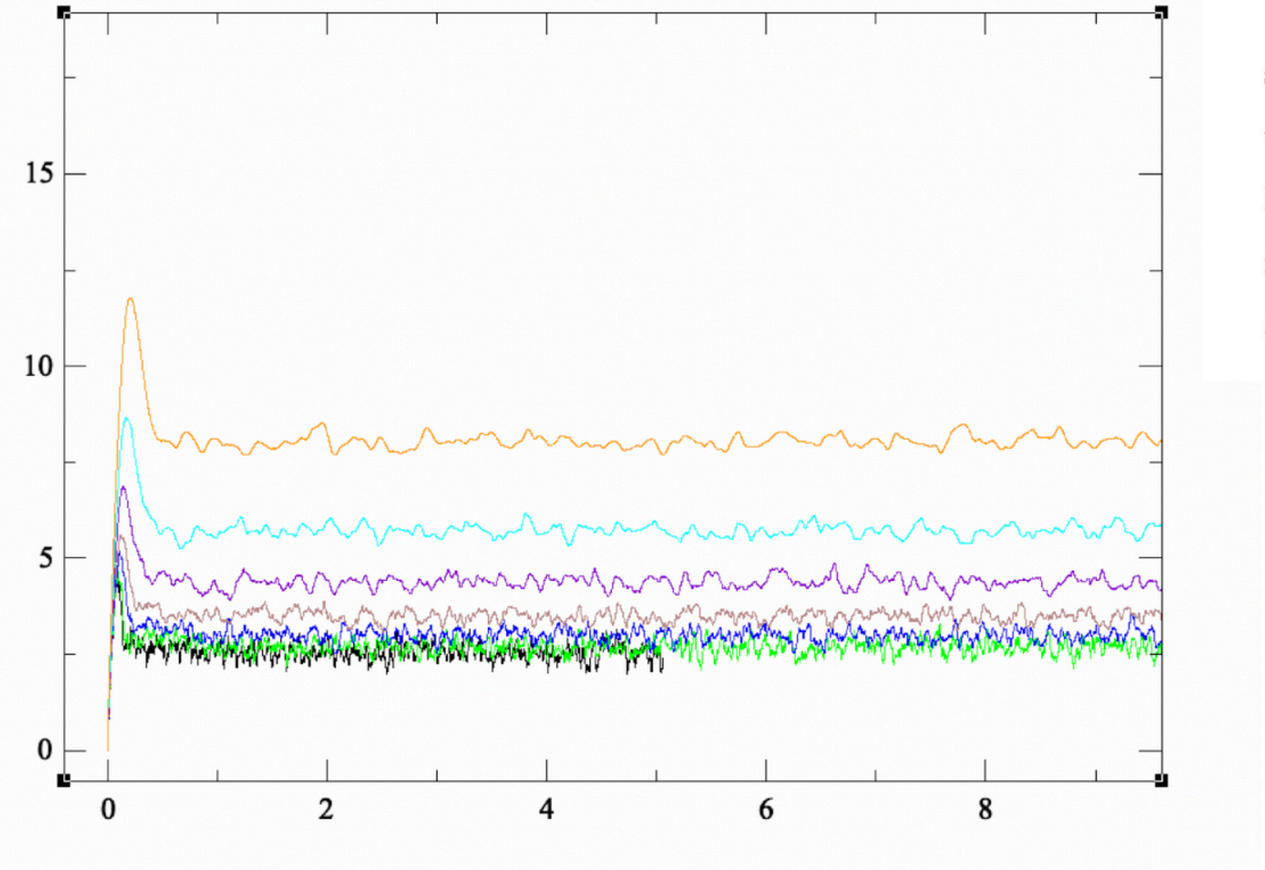
SHEAR RATES

OP1
OPO3
OPO1
OPOO3
OPOO1
OPOOO3
OPOOO1

THE RESULTS FROM MY EXAMPLES WERE THE EXACT OPPOSITE FROM THE PREVIOUS RESEARCH

POSSIBILITIES
HUMAN ERROR IN RESEARCH
REACTION IN TEMPERATURE OR SHEAR
DENSITY AND TIME APART

CONFIGURATION 2



SHEAR RATES

OPO3

OPO1

OPOO3

OPOO1

OPOOO3

OPOOO1

On the other hand, the curves in Figure 11b illustrate the shear stress with respect to strain for very well-aged sample, prepared at a cooling rate of $5 \times 10^{-6}[\varepsilon/k_B\tau_0]$, sheared at various rates. One key observation to note is that, like the poorly aged samples shown in Figure 11a, the peak stress value reached by the system also increases as the imposed shear rate is increased. However, unlike for the poorly aged sample, these load curves demonstrate an increasing abruptness in the stress drop, S_{max}, as the shear rate decreases. The curve

Insights & Future

- Differences in research
- More detailed research of examples in modern engineering and physics
- Potential to aid in environment stability and aiding climate change and chemical waste

